A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 36

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Many different organic substances may be present, each with a different potential to cause deoxygenation, so it is difficult to measure or forecast them individually **1** BOD measures the effect (amount of oxygen used by microbes decomposing the organic matter) under standardised conditions (eg 5 days, 20 °C, dark) **1** It gives a single measure directly related to the risk of deoxygenation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Biochemical Oxygen Demand** → the oxygen used by aerobic decomposers in a water sample • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'a high BOD means the water contains a lot of oxygen' - a high BOD means a lot of oxygen will be removed	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** E. coli is a common gut bacterium so its presence confirms faecal contamination and its number indicates how serious the contamination is **1** Pathogens are not always present so their absence would not prove the water is uncontaminated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Different species have different sensitivities to pollution **1** Easy to identify **1** Easy to find / normally present / usually common **1** Widely distributed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.4	**1 mark each** 1 mark for each of the 2 points below. **1 Advantage:** assesses the effect of pollution over a period of time (recent pollution events are detected even if the pollutant has gone) – no expensive equipment needed **1 Disadvantage:** identification needs skill – does not identify which pollutant is present so further chemical tests are required – species may be absent for other reasons **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 273-275; AQA spec 3.4.3.2.9.	
02.1	**1 mark each** 1 mark for each of the 2 points below. **1** The void fills with water naturally – wetlands are uncommon / valuable habitats in most regions **1** The new habitat may have greater wildlife value than the farmland lost – low cost compared with filling and levelling **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1 Housing:** ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) **1 Farmland:** landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	1 mark for the point below. **1** Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost – transplanted plants often fail **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Nuclei of large atoms (uranium-235, plutonium-239) split when hit by a neutron **1** A small amount of mass is converted to a large amount of energy ($E = mc^2$) and more neutrons are released, causing a chain reaction (controlled by control rods / moderator) **1** The heat boils water to steam which drives turbines and generators **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **control rods** → absorb neutrons to control the rate of fission • **moderator** → slows the neutrons so that fission continues **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	3
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Output changes slowly (days) so it cannot follow rapid fluctuations in demand **1** Very high construction cost but low fuel cost means it is most economic running continuously at full output **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Only 30 t of fuel a year (very high energy density) so no need for railways / ports to deliver millions of tonnes of coal **1** Still need condenser cooling water (coast / large lake) and a stable site **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
03.4	1 mark for the point below. **1** Uranium must be mined, purified, concentrated and processed into fuel using (fossil) energy – reactor construction and decommissioning have embodied energy **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **decommissioning** → safely dismantling a power station at the end of its life • **embodied energy** → all the energy used to make and transport a product **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	1
03.5	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** No commercial reactor has been fully decommissioned so costs are unknown and often exceed estimates **1** Large projects often overrun original cost estimates **1** Long-term waste disposal costs uncertain **1** State subsidies may hide costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 191-193; AQA spec 3.3.4.1.	
04.1	1 mark for the point below. **1** The time taken for half of the atoms of a radioactive isotope to decay (activity halves) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	1

Do not write outside the box

Question	Marking guidance	Marks
04.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 32 / 8 = 4 half-lives **1** 6400 / 2^4 = 400 Bq **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (Bq) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** Short half-life isotopes (I-131) release all their radiation quickly: dangerous but only briefly so short-term precautions (potassium iodide tablets, evacuation) are sufficient **1** Very long half-life isotopes (U-238) emit only small amounts of radiation in any period so the level of danger is low **1** Decades-long half-lives combine a high rate of radiation release with persistence for most or all of a human lifetime, so short-term protection is inadequate (land contaminated for generations) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	3
04.4	**1 mark each** 1 mark for each of the 3 points below. **1** Alpha particles have very low penetrating power and are absorbed by clothing / dead skin so external sources cause little exposure **1** Inside the body the alpha particles are absorbed in a very small mass of tissue so damage is concentrated (high RBE - weighting factor 20) **1** Long half-life means it continues to emit for the rest of the person's life / cannot be excreted quickly (contamination not just exposure) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	3

Do not write outside the box

Question	Marking guidance	Marks
04.5	1 mark for the point below. **1** A previously stable nucleus that becomes radioactive after absorbing a neutron (eg in reactor structures), later decaying to emit ionising radiation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 299-301; AQA spec 3.4.3.2.14.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Aerobic decomposition produces little methane and returns nutrients to the soil	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** It may contain concentrated heavy metals / toxic residues **1** It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
05.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Impermeable polymer / clay liner to stop leachate escaping **1** Collection and treatment of leachate **1** Collection of methane and use as fuel **1** Daily covering of waste / final capping and restoration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Impermeable** → does not let water through • **leachate** → polluted liquid draining out of waste **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** Quadrat size depends on the size / spacing of the organisms: 0.5 m x 0.5 m is typical for ground flora **1** 10 cm x 10 cm is for lichens (too small to contain a representative sample of plants) – 100 m x 100 m is for trees (impractical, too much work) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area	2
06.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** N = 240 – N(N-1) = 240 x 239 = 57 360 **1** Σn(n-1) = 48x47 + 120x119 + 36x35 + 24x23 + 12x11 = 2256 + 14280 + 1260 + 552 + 132 = 18 480 **1** D = 57 360 / 18 480 = 3.10 (accept 3.1) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Simpson** → a diversity index that uses both the number of species and their abundance • **Biodiversity** → the variety of life: habitat, species and genetic diversity **Do not credit** • A bare wrong answer with no working. • 'more species always means a higher index' - the evenness of the abundances matters as well	3
06.3	**1 mark each** 1 mark for each of the 2 points below. **1 Lower D means lower biodiversity:** the ungrazed meadow is dominated by one or a few species (eg tall grass) with the others rare (less even) **1** Grazing (plagioclimax management) prevents dominance by vigorous species so abundance is more evenly spread **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **plagioclimax** → a community held back from its climax by human activity	2

Do not write outside the box

Question	Marking guidance	Marks
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** A frame quadrat flattens tall plants, making percentage cover estimates inaccurate and hiding shorter plants **1** Point quadrat pins are lowered through the vegetation – each touch counts (10 pins = 10% each) without flattening **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area	2
06.5	1 mark for the point below. **1** Subjective judgement varies between recorders – results are not quantitative so statistical analysis is not possible (but quick for many areas) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-395, 401-402; AQA spec 3.7.2.1.	
07.1	**1 mark each** 1 mark for each of the 5 points below. **1** No large dam / reservoir needed so no flooding of farmland, homes or habitats and no barrier across the whole river (run-of-river / micro-hydro diverts part of the flow) **1** Suitable for rivers with a small drop in height and lower flow rates that are common in lowland rural areas **1** Small scale matches local demand – electricity used locally without long transmission lines / grid connection **1** Low construction cost and simple technology maintained by the community **1** Helical turbines tolerate high turbidity / debris without damage and fish pass through unharmed – Kaplan turbines with adjustable blades cope with variable flow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	5
07.2	1 mark for the point below. **1** Lack of capital / finance, technical knowledge or grid access – seasonal drying of rivers – conflicts with irrigation water – planning / land ownership **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
07.3	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Output is proportional to flow x efficiency: 100 x 0.10 x (40 / 90) **1** = 4.4 kW (accept 4.4-4.5) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (kW) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
07.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Constant regulated flow removes seasonal extremes: dry-season sandbanks used by nesting birds / turtles lost – high flows that clear gravel spawning beds lost – floodplain no longer flooded **1** Sudden flow increases for peak generation raise turbidity, wash away species, reduce light for photosynthesis **1** Sediment trapped so less nutrient replenishment downstream – barrier to migrating fish (salmon, sturgeon) and isolation of populations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 203-205; AQA spec 3.3.4.1.	
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** B, C and D – each loses soil faster than the 1 t ha^{-1} yr^{-1} at which it forms, so the soil is getting shallower **1** Accelerated erosion means the natural equilibrium between soil formation and erosion has been broken – field A loses only 0.8 t ha^{-1} yr^{-1}, which is less than the rate of formation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	2

Do not write outside the box

Question	Marking guidance	Marks
08.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (24.0 - 9.0) / 24.0 x 100 **1** = 62.5% (accept 62 - 63%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **contour ploughing** → ploughing along the slope rather than up and down it **Do not credit** • A bare wrong answer with no working.	2
08.3	**1 mark each** 1 mark for each of the 3 points below. **1 Contour ploughing:** furrows run along the contour at 90° to the slope so runoff is stopped by the furrow and loses kinetic energy – the particles it was carrying are deposited in the furrow instead of being washed away **1 Terracing:** the slope is cut into a series of narrow level fields held by retaining walls, so water slows as it crosses each field and much of it infiltrates rather than running off **1 Rows of stones:** stones laid along the contours of a gentle slope slow the runoff and trap disturbed soil behind them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Contour ploughing** → ploughing along the slope rather than up and down it • **Terracing** → cutting the slope into level steps	3

Do not write outside the box

Question	Marking guidance	Marks
08.4	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1 Multicropping:** growing more than one crop in the field at a time, sown and harvested at different times, so some crop always covers the soil **1 Strip cropping:** wide strips of different crops, most effective at 90° to the prevailing wind, for farms that use machinery **1** Growing a long-term crop such as permanent grassland or a bush or tree crop, which does not need frequent replanting or soil disturbance **1** No-tillage cultivation / direct drilling of seed, which disturbs the soil far less than ploughing and sowing **1** Adding organic matter such as mulch or manure – surface mulch shields the soil from the wind and from rain splash, and the humus formed as it decomposes makes the particles adhere to each other **1 Controlling livestock:** lower stocking density and restricted movement so there is less trampling damage and more plant cover **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
08.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Soil particles carried into rivers are deposited where the water slows, so the flow capacity of the river falls and it overflows more readily after heavy rain **1** Sediment fills reservoirs, reducing the volume of water they can store and shortening their useful life **1** High turbidity in the river reduces light penetration and smothers the river bed, damaging aquatic life **1** Wind erosion raises the level of atmospheric particulates, which worsens respiratory illness such as asthma **1 Erosion contributes to desertification:** vegetation struggles to grow on the thinner soil, which exposes more soil and reduces rainfall further **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **particulates** → fine solid particles / smoke and dust **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-166; AQA spec 3.2.5.2.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**Marks shared** Up to 6 marks in total - share them between the 5 points below according to how completely each one is made. **up to 2 Habitat change:** flooding destroys the previous terrestrial / river habitats but creates a wetland which may be valuable (wetlands are uncommon) **up to 2 Wildlife barrier:** the dam blocks migrating fish (salmon, sturgeon) and prevents recolonisation along the river **up to 2 Changed river regime downstream:** regulated flow removes low-flow periods needed by species such as river turtles nesting on sandbanks and high flows that wash sediment from gravel spawning beds **up to 2 Sedimentation:** sediment settles in the reservoir instead of fertilising the floodplain and building river banks / coastlines downstream, increasing erosion **up to 2 Microclimate:** water's high heat capacity reduces temperature fluctuations – higher wind speeds – more evaporation, humidity, cloud and precipitation downwind **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	6
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Vegetation / dead organic matter under water decays anaerobically **1** Releasing methane, a powerful greenhouse gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** After heavy rain surplus water is stored, reducing peak flow / flooding downstream **1** In dry periods the dam releases water to maintain river levels / supplies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117; AQA spec 3.2.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Features:** renewable / non-renewable, depletable, abundance, energy density, locational constraints, intermittency and predictability, need for conversions, applicability, ease of storage and transport, technological development, political and economic influences, environmental impacts • **Fossil fuels:** high energy density, easy storage and transport, reach high temperatures, well-developed technologies and infrastructure - but finite, polluting (acid rain, smoke, CO2), politically concentrated (Middle East oil) • **Nuclear:** very high energy density, base load, few fuel deliveries - but complex, expensive, waste, public opposition • **Renewables:** low energy density, intermittent (wind, solar unpredictable; tidal predictable), locationally constrained, cannot be stored without conversion, need storage / grid, cannot reach fossil-fuel temperatures or replace liquid vehicle fuels yet - but renewable, low pollution, falling costs • **Sustainability concepts:** impacts before use (extraction, processing, equipment embodied energy) and as a consequence of use (pollution, habitat damage, depletion) • **Economic and political factors:** subsidies, guaranteed prices, planning, external costs not included in price, energy security • **Judgement:** a diverse mix matched to uses (electricity vs heat vs transport) with storage is most sustainable – transition must start before depletion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **sustainability** → meeting present needs without stopping future generations meeting theirs • **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous • **base load** → the steady minimum demand that must always be met **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (8 marks)	
	Whether a resource suits a use depends on its features: renewable or not, abundance, energy density, locational constraints, intermittency and predictability, the conversions needed, ease of storage and transport, how developed the technology is and its impacts. Fossil fuels dominate because they combine high energy density, easy storage and transport, high temperatures for industry and mature infrastructure - which is why petrol and diesel still monopolise transport. But they are finite, cause acid rain, smoke and CO_2, and are politically concentrated. Nuclear fuel has the highest energy density and gives reliable base load with few deliveries, yet the technology is complex and expensive and produces long-lived waste against public opposition.	
	Renewables have the opposite profile: low energy density needing large collectors, intermittency (wind and solar unpredictable, tidal predictable), strong locational constraints and no direct storage, so they need conversion, storage and grid links and cannot yet reach fossil-fuel temperatures or replace liquid fuels. Sustainability must count impacts before use - extraction, processing, embodied energy of equipment - as well as pollution, habitat damage and depletion during use, and economics and politics (subsidies, planning, external costs left out of prices, energy security) shape what is actually chosen.	
	No single resource is suitable for everything. Electricity, heat and transport have different requirements, so the most sustainable system is a diverse mix matched to each use with storage, and the transition must begin before depletion forces it.	
	Weaker answer - Level 1 (2 marks)	
	Energy resources have different features. Fossil fuels are non-renewable and will run out but they have a high energy density. Renewable energy like wind and solar will not run out but they are intermittent because the wind does not always blow. Nuclear power produces radioactive waste. Renewable energy is more sustainable because it does not produce carbon dioxide. So we should use more renewable energy.	
	What makes the difference	
	• Level 1 mentions two or three features; Level 3 works through the full set and links each to a specific use (transport, base load, industry).	
	• Level 3 includes sustainability concepts (impacts before and during use, external costs) and political factors that Level 1 omits entirely.	
	• The conclusion in Level 3 is a reasoned position (a diverse mix matched to uses); Level 1's 'use more renewables' does not follow from a discussion.	

Do not write outside the box

Question	Marking guidance	Marks
10.2	1 mark for the point below. **1** Very high energy density fuel means only a few tens of tonnes are needed each year, compared with 10 000 t of coal per day for a coal-fired station **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-182; AQA spec 3.3.2-3.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Scale of the problem:** how energy is used (transport, buildings, industry, water) – links between affluence and per capita use • **Transport:** bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • **Buildings:** orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • **Industry:** heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • **Limits:** rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples – comparison with the cost of new supply	25
	– conclusion that conservation is the cheapest first step but cannot alone meet demand growth *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Context:** uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • **New sources:** rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • **Management:** artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • **Conservation:** low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering – pollution control to protect sources • **Treatment:** matching quality to use – screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	

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